

Mini Project 1

SYNTHESIS OF UREA AS A NITROGENOUS FERTILIZER

*Design Analysis of an Integrated Ammonia-Urea Production Plant Utilizing the
Haber-Bosch and Stamicarbon CO₂ Stripping Processes*

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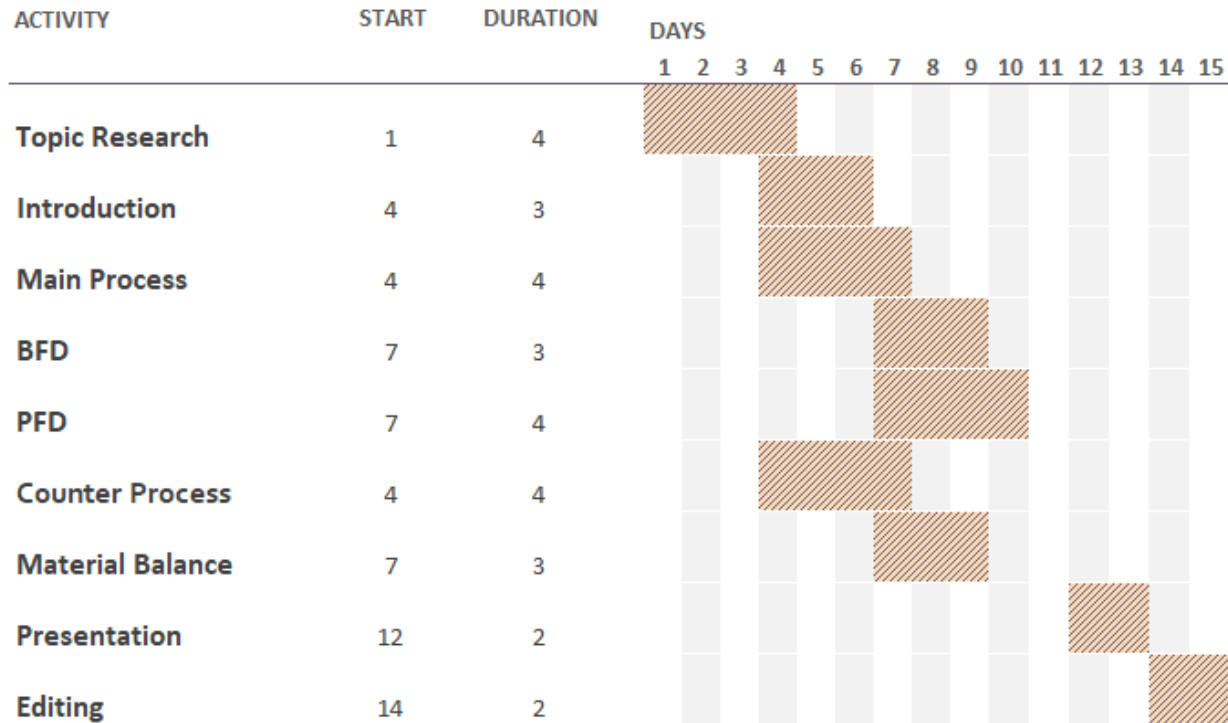
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Project Charter

Our project considers the production of an integrated ammonia and urea production plant using the Haber-Bosch and Stamicarbon CO₂ Stripping processes. The purpose of this project is to design a safe, efficient, and cost-effective integrated ammonia-urea facility that produces fertilizer-grade urea. This project aims to evaluate the current industry processes and replicate them, developing our own process flow diagrams and mass and energy balances. By analyzing the current processes, this project will help highlight the importance of optimization and investigate more sustainable methods for the future.

The project scope and deliverables include setting the design basis, which consists of establishing the size and location of the plant, defining the raw materials needed, and drawing block flow and process flow diagrams. The block flow diagram will reflect the major steps of the process, while the preliminary process flow diagram will include the major equipment necessary for our facility. We will also prepare a rough material balance that considers the various inputs and outputs of the process blocks. The balances should include estimates for raw material quantities based on the chosen size of the plant. The primary deliverables for our group will be a written report explaining the design as well as a final presentation to present our objectives, process, and conclusions.

Design Plan



Introduction

Problem Definition and Significance

Global agriculture is heavily dependent on synthetic fertilizers to sustain crop yields and feed a growing population. Two of the most important components of nitrogen-based fertilizers are ammonia (NH_3) and urea ($\text{CO}(\text{NH}_2)_2$), which together supply most of the plant-available nitrogen worldwide. Therefore, ammonia and urea production are cornerstones of the global food supply chain. Roughly 80% of all ammonia produced is used as fertilizer, directly or as a precursor to urea, ammonium nitrate, and other nitrogen fertilizers. Urea is particularly important due to its high nitrogen content (46% by weight), low cost per unit of nitrogen, and ease of handling. With the

world's population steadily increasing, nitrogen demand is expected to continue increasing as well, making reliable and efficient fertilizer production essential for food security.

Beyond that, ammonia and urea production have global sustainability implications. Advances such as low-carbon hydrogen production, improved catalysts, and energy-efficient CO₂ stripping technologies can significantly reduce greenhouse gas emissions

Process Description

Ammonia and urea production are among the most important chemical processes in the world, necessary to sustain global agriculture. In this project, we will analyze how ammonia is traditionally synthesized via the Haber–Bosch process, which combines nitrogen and hydrogen at high pressure and temperature using iron-based catalysts. Urea is then produced by reacting ammonia with carbon dioxide in a high-pressure loop, using a process known as the Stamicarbon CO₂ Stripping process. We chose these as our main processes as they are the dominant technologies used in industry today.

Hydrogen Production

The front end of our ammonia-urea plant begins with the production of hydrogen. This section begins with the desulfurization of natural gas, where sulfur compounds in the feed are removed using a zinc oxide bed in order to protect the nickel catalyst used in the next reaction. The desulfurized methane is then mixed with steam and sent to a Steam Methane Reformer (SMR), where it passes over a nickel catalyst at 800–900 °C and moderate pressure (20–30 bar). The heat required for the reforming reaction is supplied by a fired furnace, and the hot effluent gas is used to generate high-pressure steam in waste-heat boilers before further processing. In this highly

endothermic process, methane reacts with steam to produce a synthetic gas (syngas) containing hydrogen, carbon monoxide, carbon dioxide, and water vapor.

The syngas is then sent to a secondary SMR where it is mixed with a stoichiometrically calculated amount of air. This results in exothermic combustion where oxygen and excess methane produce more hydrogen. This completes the conversion of methane and also introduces nitrogen into the gas mixture in the exact stoichiometric proportion required for ammonia synthesis ($\text{H}_2:\text{N}_2 \sim 3:1$), which eliminates the need for a separate air separation unit. This gas mixture is cooled before moving on to the Water–Gas Shift (WGS) reactor. The addition of air results in a small amount of inert argon, which is purged and can be treated in a purge gas recovery unit to recover hydrogen and nitrogen for recycling, while the argon off-gas can be used as fuel for the reformer furnace.

The steam generated is an important part of the plant's steam system, which recycles and distributes high-pressure steam throughout the facility. The steam produced in the SMR is used to drive many of the other processes in the plant, including the CO_2 stripping column in the urea loop, feed and recycle compressors, and other steam-heated exchangers, improving overall energy efficiency by recovering heat that would otherwise be wasted.

After steam generation, the reformer effluent enters the WSG reactor, where carbon monoxide reacts with additional steam to form carbon dioxide and more hydrogen. This mildly exothermic reaction maximizes hydrogen yield while producing a concentrated CO_2 stream for use in the urea synthesis plant. The gas mixture is then passed through a CO_2 removal column. This results in an overhead product of a CO_2 -rich stream for urea production and a bottoms product of hydrogen and nitrogen. Any remaining traces of CO or CO_2 are converted to methane in the methanation reactor to protect the iron ammonia synthesis catalyst used in the Haber-Bosch

process, and the methane is recycled to the SMR. The resulting gas is a clean hydrogen/nitrogen mixture in the stoichiometric 3:1 ratio required for the Haber–Bosch ammonia synthesis loop.

Ammonia Production

The ammonia production takes the hydrogen/nitrogen mixture from the front end and compresses it to the high pressure required for the Haber-Bosch synthesis loop, typically 150-250 bar. This compressed gas is fed to the ammonia synthesis reactor, a multibed reactor containing an iron catalyst. In the reactor, nitrogen and hydrogen are converted to ammonia in an exothermic reaction. Operating temperatures are maintained at 400-500 °C to keep the equilibrium conversion favorable. The effluent is composed of ammonia along with unreacted hydrogen and nitrogen. It is cooled, and the liquid ammonia is separated and sent to a tank for the urea synthesis section. The unconverted hydrogen and nitrogen gases are recycled back into the ammonia synthesis reactor using a recycle compressor, which allows for higher conversion of the reactants. This recycle loop also has waste-heat boilers that recover heat to generate high-pressure steam for the plant's steam system. As mentioned in the hydrogen production section, this steam is used elsewhere in the facility, improving the overall thermal efficiency of the plant. To prevent inert gas accumulation that was introduced with process air, a small purge stream can be used, similar to the one needed after the WGS. This closed-loop design ensures maximum ammonia yield and minimizes energy waste.

Urea Production

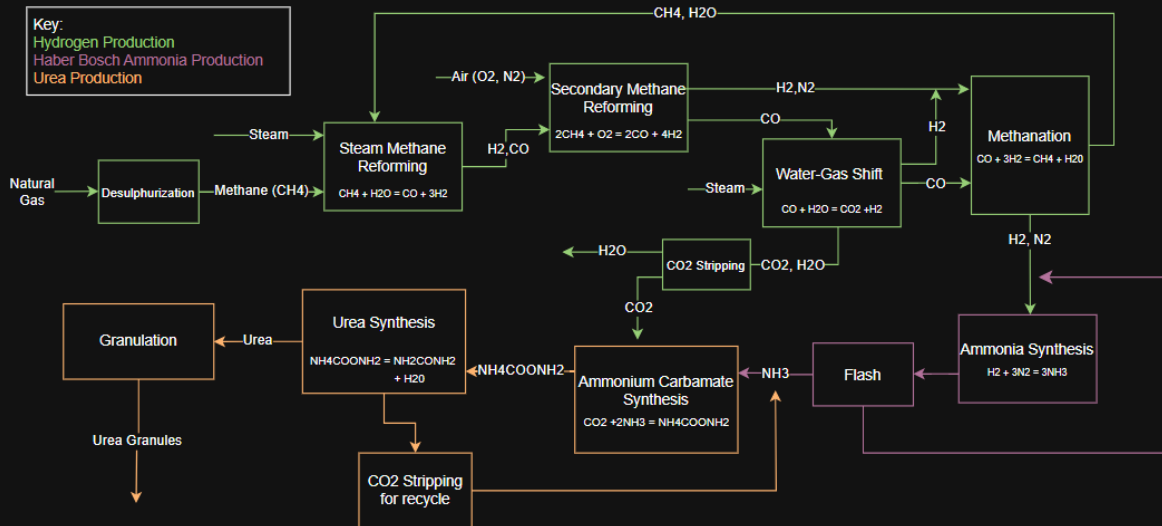
The urea production takes liquid ammonia from the ammonia synthesis loop and carbon dioxide recovered from the hydrogen production and feeds them into a high-pressure urea synthesis reactor. Inside the reactor, ammonia and carbon dioxide form urea in a two-step reaction.

First, ammonium carbamate (NH_2CONH_4) is rapidly produced in a highly exothermic reaction. Then the ammonium carbamate is dehydrated under high pressure in a slower, endothermic reaction. The reactor operates at 140 bar and 180-190 °C to maximize conversion.

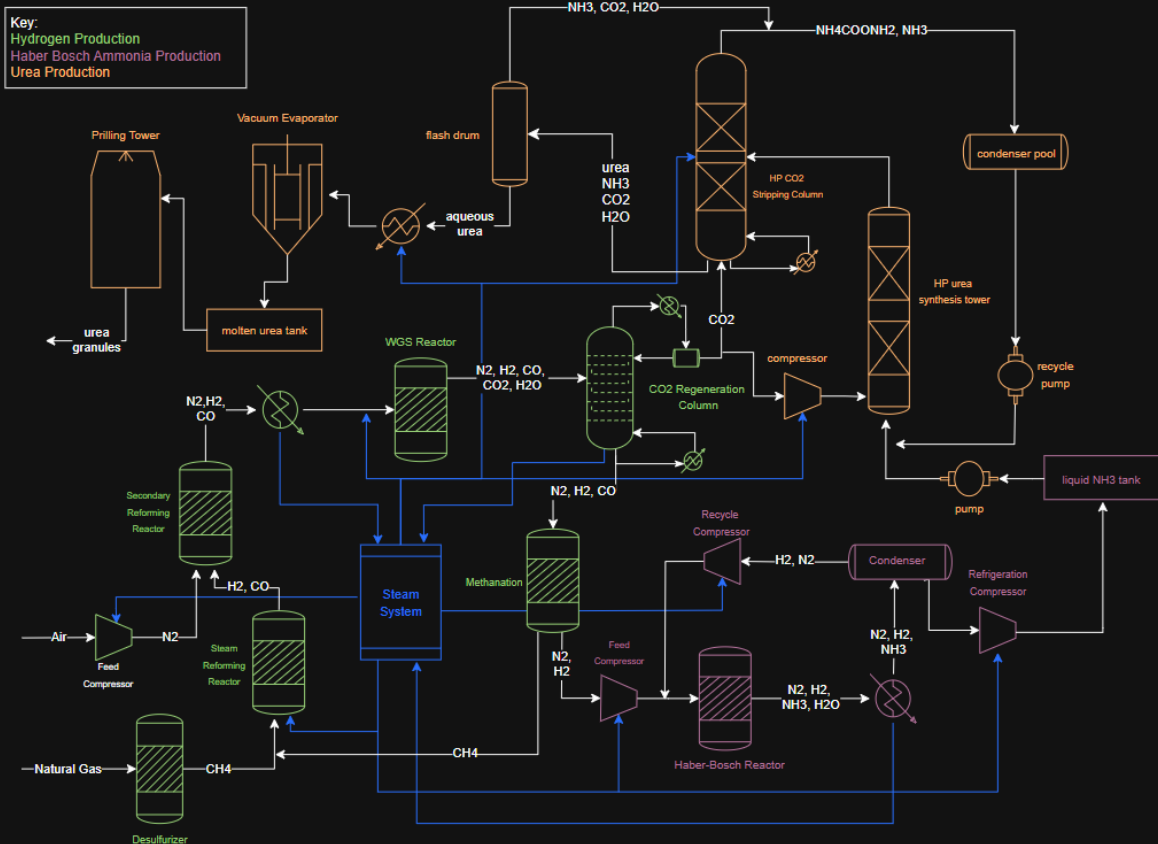
To improve yield and maximize energy efficiency, we utilize the Stamicarbon CO_2 stripping process. In this process, the mixture leaving the reactor, which contains urea, water, unconverted carbamate, and ammonia, is passed through a CO_2 stripping column. Here, heat and a countercurrent of CO_2 strip unconverted ammonia and carbamate out of the liquid. These gases are sent to a condenser pool, forming a recycle stream that is pumped back into the reactor at high pressure, creating a closed synthesis loop with high overall conversion. As in the previous sections, the heat released during the carbamate condensation can be used to generate steam to power other process needs.

The urea-rich solution leaving the stripper is sent to a flash drum, where the sudden pressure drop from 140 bar to 15 bar results in additional ammonia and carbon dioxide being removed as gas, which is then condensed and recycled. The liquid phase of aqueous urea leaves the bottom of the drum and is then concentrated in a vacuum evaporator, reducing the water content. The molten urea is then sent to a prilling tower or granulation unit, as urea fertilizer is sold in 2 – 4 mm diameter pellets. The molten urea is sprayed from the top of a tall tower through a perforated plate, breaking the liquid into thousands of tiny droplets which fall through the tower. Air is blown countercurrent to the falling droplets, causing them to cool and solidify into nearly spherical urea prills. These solid prills are collected at the base of the tower and cooled further before they are transported to packaging for shipment.

Block Flow Diagram

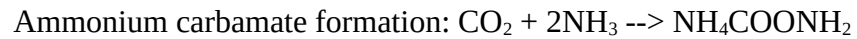
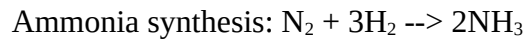
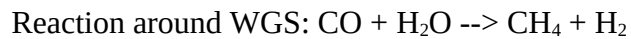


Process Flow Diagram



Material Balances

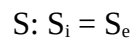
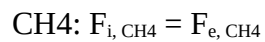
All reactions



Block Balances

(ξ is extent of reaction)

Desulphurization



SMR (S: Steam)

$$\text{CH}_4: F_{i, \text{CH}_4} + F_{iM, \text{CH}_4} = F_{e, \text{CH}_4} + \xi_1$$

$$\text{H}_2\text{O}: S_{\text{SMR}} + F_{iM, \text{H}_2\text{O}} = F_{e, \text{H}_2\text{O}} + \xi_1$$

$$\text{CO}: \xi_1 = F_{e, \text{CO}}$$

$$\text{H}_2: 3\xi_1 = F_{e, \text{H}_2}$$

Secondary Methane Reformer (A: Air)

$$\text{CH}_4: F_{i, \text{CH}_4} = F_{e, \text{CH}_4} + 2\xi_2$$

$$\text{O}_2: A_{\text{O}_2} = F_{\text{O}_2} + \xi_2$$

$$\text{N}_2: A_{\text{N}_2} = F_{\text{N}_2}$$

$$\text{CO}: F_{i, \text{CO}} + 2\xi_2 = F_{e, \text{CO}}$$

$$\text{H}_2: F_{i, \text{H}_2} + 4\xi_2 = F_{e, \text{H}_2}$$

$$\text{H}_2\text{O}: F_{i, \text{H}_2\text{O}} = F_{e, \text{H}_2\text{O}}$$

WGS

$$\text{CO}: F_{i, \text{CO}} = F_{e, \text{CO}} + \xi_3$$

$$\text{H}_2\text{O}: F_{i, \text{H}_2\text{O}} = F_{e, \text{H}_2\text{O}} + \xi_3$$

$$\text{CO}_2: F_{i, \text{CO}_2} + \xi_3 = F_{e, \text{CO}_2}$$

$$\text{H}_2: F_{i, \text{H}_2} + \xi_3 = F_{e, \text{H}_2}$$

Stripper

$$Y_{\text{CO}_2} F_{i, \text{CO}_2} = S_{\text{stripped, CO}_2}$$

$$(1 - Y_{\text{CO}_2}) F_{i, \text{CO}_2} = F_{e, \text{CO}_2}$$

$$F_{i, \text{H}_2\text{O}} = F_{e, \text{H}_2\text{O}}$$

Methanation

$$\text{CO}: F_{i, \text{CO}} = F_{e, \text{CO}} + \xi_4$$

$$\text{H}_2: F_{i, \text{H}_2} = F_{e, \text{H}_2} + 3\xi_4$$

$$\text{CH}_4: F_{i, \text{CH}_4} + \xi_4 = F_{e, \text{CH}_4}$$

$$\text{H}_2\text{O}: F_{i, \text{H}_2\text{O}} + \xi_4 = F_{e, \text{H}_2\text{O}}$$

Note: Some CH_4 and H_2O gets recycled to SMR

Ammonia Synthesis

$$\text{N}_2: F_{i, \text{N}_2} = F_{e, \text{N}_2} + \xi_5$$

$$\text{H}_2: F_{i, \text{H}_2} = F_{e, \text{H}_2} + 3\xi_5$$

$$\text{NH}_3: F_{i, \text{NH}_3} + 2\xi_5 = F_{e, \text{NH}_3}$$

Flash

$$S_{\text{NH}_3} F_{\text{e, NH}_3} = F_{\text{to urea, NH}_3}$$

$$F_{\text{recycle, H}_2} = r_{\text{recycle}} F_{\text{out, H}_2}$$

$$F_{\text{recycle, N}_2} = r_{\text{recycle}} F_{\text{out, N}_2}$$

Note: Recycled back to ammonia synthesis

Ammonium Carbamate Synthesis

$$\text{CO}_2: F_{\text{i, CO}_2} = F_{\text{e, CO}_2} + \xi_6$$

$$\text{NH}_3: F_{\text{i, NH}_3} = F_{\text{e, NH}_3} + \xi_6$$

$$\text{NH}_4\text{COONH}_2: F_{\text{i, NH}_4\text{COONH}_2} + \xi_6 = F_{\text{e, NH}_4\text{COONH}_2}$$

Urea Synthesis

$$\text{NH}_4\text{COONH}_2: F_{\text{i, NH}_4\text{COONH}_2} = F_{\text{e, NH}_4\text{COONH}_2} + \xi_7$$

$$\text{NH}_2\text{CONH}_2: F_{\text{i, NH}_2\text{CONH}_2} + \xi_7 = F_{\text{e, NH}_2\text{CONH}_2}$$

$$\text{H}_2\text{O}: F_{\text{i, H}_2\text{O}} + \xi_7 = F_{\text{e, H}_2\text{O}}$$

CO₂ Stripping

$$R_{\text{CO}_2} F_{i, \text{CO}_2} = F_{e, \text{CO}_2}$$

Note: This is fed to reaction 6

Granulation

$$F_{i, \text{NH}_2\text{CONH}_2} = F_{e, \text{NH}_2\text{CONH}_2}$$

Plant basis

Our proposed plant is based on QAFCO's fertilizer plant, which is in Qatar. It produces 3.8 million MT of ammonia and 5.6 million MT of urea annually. Due to the substantial annual production, QAFCO is the world's largest single site producer of ammonia and urea. In turn, it made Qatar the 4th largest urea producer. Qatar is an optimal location for our plant since it is close to natural gas sources, which is needed to produce ammonia, and by extension, urea. The natural gas that would be used by our plant would be managed by Ras Laffan Industrial City, which has well-developed industrial infrastructure, suitable for natural gas. Additionally, Qatar is the largest proven natural gas reserves in the world. Furthermore, Qatar is an ideal location for trade, due to its proximity to major urea fertilizer markets such as India, China, SE Asia, Africa, and Europe. This makes trade unchallenging. Lastly, like our process, QAFCO opts for granular urea over urea pills. Granular urea is more situated for developed countries with technology dependent cultivation methods.

Calculations

$$\text{Moles NH}_3 = \frac{3.8\text{E6 ton}}{17.031 \text{ g/mol}} \times 10\text{E6 mol/year} = \mathbf{2.231\text{E11 moles NH}_3/\text{year}}$$

$$\text{Moles Urea} = \frac{5.6\text{E6 ton}}{60.056 \text{ g/mol}} \times 10\text{E6 mol/year} = \mathbf{9.325\text{E10 moles urea/year}}$$

Feed Requirements:

$$4n_{\text{CH}_4} = 3n_{\text{urea}} + 1.5n_{\text{NH}_3} \rightarrow \text{moles CH}_4 = 1.536\text{E11 moles/year} \times 16.042 \text{ g/mol} = \mathbf{2.464\text{E6 tons natural gas/year}}$$

$$\text{Moles N}_2 = \text{Moles Urea} + 0.5 \text{ Moles Ammonia}$$

$$9.325\text{E10} + 0.5 \times 2.231\text{E11} = 2.046\text{E11 moles N}_2/\text{year} \times 26.014 \text{ g/mol} = 5.737\text{E6 tons N}_2/\text{year}$$

Air required: Assuming air is 79 molar percentage N₂.

$$\text{Moles of N}_2 / 0.79 = \mathbf{2.59\text{E11 moles air/year}}$$

CO₂ Production:

$$1 \text{ mol CO}_2 \text{ produced for 1 mol CH}_4 \text{ reformed} \rightarrow 1.536\text{E11 moles CO}_2 / \text{year} \times 44.01 \text{ g/mol} = \mathbf{6.76\text{E6 tons CO}_2/\text{year}}$$

To synthesize 1 mole of ammonium carbamate and make 1 mole of urea, 1 mole of CO₂ is used. In other terms:

Moles CO₂ used: $9.325\text{E}10 \text{ moles CO}_2/\text{year} \times 44.01 \text{ g/mol} = 4.104\text{E}6 \text{ tons CO}_2/\text{year}$

Excess CO₂ → $1.536\text{E}11 - 9.325\text{E}10 = 6.036\text{E}10 \text{ moles CO}_2/\text{year} \times 44.01 = 2.65\text{E}6 \text{ tons of CO}_2/\text{year}$

2.65E6 tons of CO₂ that is released annually

Competing Process

Green Hydrogen

There are many innovative “green ammonia” technologies that are being researched and piloted to reduce CO₂ emissions and energy consumption. Green ammonia refers to ammonia produced without using fossil fuels as the hydrogen source, eliminating the CO₂ emissions associated with conventional steam methane reforming (SMR). In a conventional ammonia plant, natural gas is reformed with steam to produce hydrogen, releasing significant CO₂ as a by-product. This makes ammonia synthesis responsible for roughly 1–2% of global CO₂ emissions, a major contributor to industrial greenhouse gases. Green ammonia production replaces the SMR step with water electrolysis powered by renewable energy, generating hydrogen with nearly zero carbon footprint.

Interest in green ammonia is growing rapidly because it addresses two pressing global challenges: decarbonizing fertilizer production and developing a sustainable energy carrier. Fertilizer producers see green ammonia as a way to offer low-carbon nitrogen products to farmers, helping agriculture reduce its climate impact. At the same time, green ammonia is attracting attention as a zero-carbon fuel for shipping and power generation. While still more expensive than conventional ammonia, green ammonia is viewed as a critical technology pathway toward achieving net-zero emissions in both the fertilizer and energy sectors.

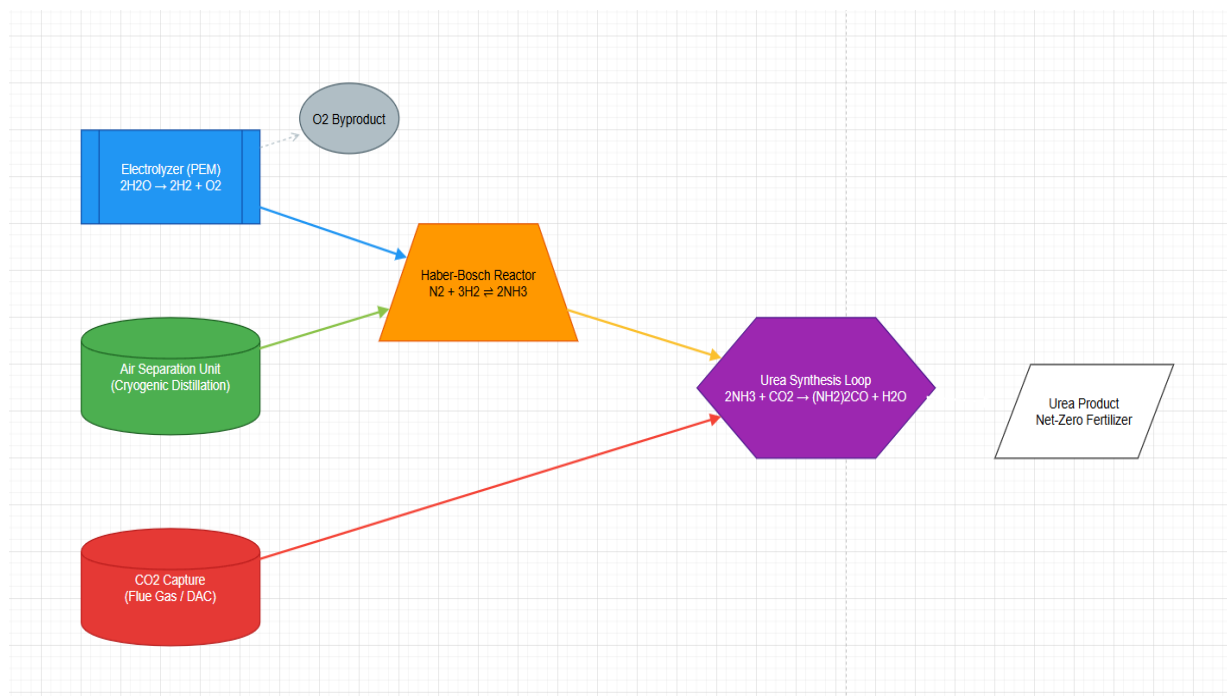
Process

Purified water is fed to a proton exchange membrane (PEM) electrolyzer, where an electric current splits water molecules into hydrogen and oxygen. PEM electrolyzers generate high-purity hydrogen under pressure, reducing the need for downstream compression. The electricity supply is sourced from renewable energy, such as wind, solar, or hydropower, allowing hydrogen generation with near-zero CO₂ emissions. This eliminates the large CO₂ footprint associated with SMR and WGS, making it a cornerstone technology for decarbonizing fertilizer production. The main challenges with this approach are the high electricity demand — which increases operating cost — the capital-intensive nature of electrolyzer systems, and the need to improve membrane durability to lower long-term maintenance costs.

In parallel, nitrogen is produced from atmospheric air using a cryogenic air separation unit (ASU). Air is first filtered and compressed, and impurities such as moisture, hydrocarbons, and CO₂ are removed to avoid freezing during cryogenic cooling. The clean air stream is cooled to cryogenic temperatures using multi-stage refrigeration cycles, allowing nitrogen, oxygen, and argon to be separated based on their different boiling points. The result is a high-purity nitrogen stream, which is combined with hydrogen in the correct stoichiometric ratio for ammonia synthesis.

Hydrogen from electrolysis and nitrogen from the ASU are dried, purified, and compressed to approximately 150 bar before being mixed in the stoichiometric 3:1 H₂:N₂ ratio. This mixture is fed into a conventional Haber–Bosch ammonia synthesis loop, where it reacts over an iron-based catalyst at high temperature (400–500 °C) and pressure (150–250 bar) to form ammonia. The ammonia product is then condensed, separated, and sent to the urea synthesis loop, where it reacts with captured CO₂ under high pressure to form ammonium carbamate.

Since electrolysis does not produce CO₂ as a by-product, the CO₂ used for this reaction must be captured from another source, such as flue gas from a power plant or direct air capture. By integrating green ammonia and captured CO₂, this process eliminates fossil fuels as a feedstock, enabling the production of net-zero urea fertilizer and significantly reducing the carbon footprint of nitrogen-based fertilizers.



Market analysis

The worldwide urea market continues to expand because of rising agricultural requirements and escalating food security challenges. Fortune Business Insights (2023) reports that the market value reached USD 128.9 billion in 2023, and analysts predict it will expand to USD 160–160.7 billion by 2032. The global urea market shows consistent volume growth because production and

consumption reached 180 million tons in 2024, and analysts predict it will reach 276 million tons by 2035 at a 3.9–4.0% CAGR (ChemAnalyst, 2024). The Asia-Pacific region leads the market through its control of more than half of the global market share because China and India, along with other large agricultural nations in the region, use substantial amounts of fertilizers (ChemAnalyst, 2024). The market experiences price instability in raw materials, particularly natural gas, which fuels Haber-Bosch ammonia production, and there is rising demand for environmentally friendly "green urea production methods to decrease CO₂ emissions (MarketsandMarkets, 2024).

The calculation of urea production expenses at our facility used natural gas, water, air, and carbon dioxide as primary raw materials for Haber–Bosch and Stamicarbon processes. The production process depends on natural gas and make-up water as its only required purchased inputs because air exists freely and CO₂ is recovered from reforming operations. The Haber–Bosch and Stamicarbon processes need 16 MMBtu of natural gas to produce one ton of urea, which costs between \$47 and \$51 based on current Henry Hub prices ranging from \$2.89 to \$3.10 per MMBtu (EIA, 2025; FRED, 2025). The production of urea requires 5–13 m³ of water per ton according to Down To Earth (2019) and Fiamelda & Suprihatin (2020). The water cost adds \$4–\$10 per ton to the production expenses when using an industrial water tariff of \$0.74 per m³ (City of Bryan, 2021). The estimated production cost of urea through raw materials amounts to \$51–\$61 per ton. The yearly expenses for natural gas and water at our 5.6 million ton urea production rates would amount to \$285–\$340 million without considering operational costs or capital expenditures.

The following table presents the raw material costs for our plant to produce urea through Haber–Bosch ammonia synthesis and Stamicarbon CO₂ stripping urea production. The costs span from minimum to maximum values because natural gas prices and water usage rates show

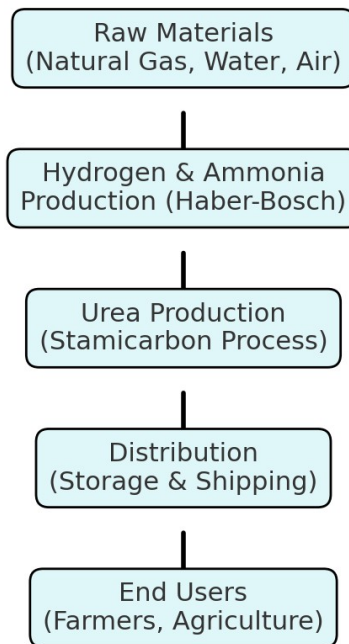
significant variation. The costs are displayed in two formats, which show both the price per ton of urea and the yearly expenses based on the plant's 5.6 million ton annual urea production.

Cost Component	Per Tonne (USD, low)	Per Tonne (USD, high)	Annual (Million USD, low)	Annual (Million USD, high)
Natural Gas	47.0	51.0	263.2	285.6
Water	4.0	10.0	22.4	56.0
Total	51.0	61.0	285.6	341.6

Supply Chain Analysis

The production of ammonia–urea fertilizer starts with the procurement of raw materials consisting of natural gas together with water and air which function as fundamental production materials. The Haber-Bosch process requires hydrogen from natural gas reforming and atmospheric nitrogen extraction to create ammonia. The Stamicarbon process receives ammonia from production to create urea through a reaction with captured carbon dioxide. The manufacturing process of urea ends with granulation or prilling before it moves through distribution channels that include storage facilities and shipping terminals and logistics networks. The final destination of urea fertilizer reaches farmers and agricultural businesses who use it to boost their crop production. The supply chain operates through physical product transportation to achieve rapid delivery from resource extraction to end-user consumption.

Supply Chain (Flow of Materials)

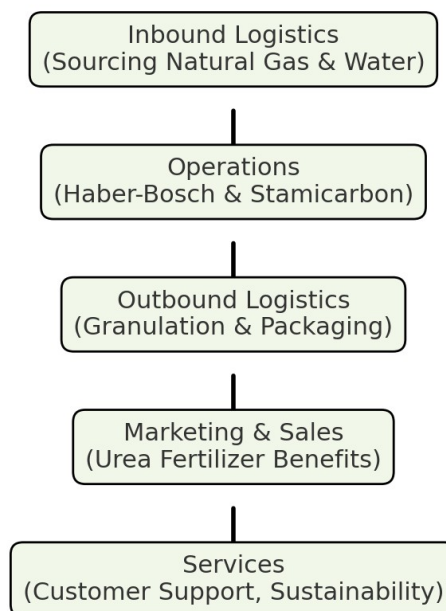


Value Chain Analysis

The value chain model examines how each stage of urea production generates competitive advantages and adds value to the product. The first stage of inbound logistics requires businesses to obtain natural gas and water resources through secure methods which provide both affordability and reliability. The Haber-Bosch and Stamicarbon processes within operations create substantial value through their optimized industrial reactions which transform basic inputs into high-demand fertilizers using high amounts of energy. The outbound logistics process enhances product value through the conversion of molten urea into granules or prills and the addition of packaging for transportation and the fulfillment of customer requirements for product handling and application. The marketing and sales efforts of the company emphasize the

advantages of urea fertilizer through its high nitrogen content and affordable price while establishing unique selling points in worldwide market competition. The combination of sustainability programs and customer service support within after-sales activities creates enduring value through environmental solutions and customer relationship development. The value chain demonstrates that business activities extending past manufacturing operations generate both financial success and customer contentment.

Value Chain (Value Creation)



Green Ammonia Analysis

The market for green ammonia is rapidly growing through global efforts to reduce carbon emissions. Despite expectations for significant growth, there are multiple challenges related to infrastructure and cost.

Key Market Statistics indicate that the market was valued between \$44.7 million to \$445.7 million in 2023-24 and is currently projected to reach \$44 billion to \$73 billion by 2030 with annual growth rates between 65% to 116% from 2024 to 2030 (Fortune Business Insights, markets and Markets). Also, the application of green ammonia is currently being used in fertilizers, there is significant traction/demand as a carbon-free fuel which can be used for energy storage and in the maritime industry (MarketsandMarkets).

Market drivers include decarbonization as governments and industries are clamoring for reductions in their emissions and enable goals for net-zero emissions. Furthermore, policies from the EU like their Renewable Energy Directive (RED III) and the US Inflation Reduction Act also give incentives to producers through subsidies (Topsoe).

Despite the positivity with progress being made, there are challenges associated with the production of green ammonia as it currently 1.5x more expensive than the traditional fossil-fuel based ammonia due to high costs of renewable energy infrastructure and electrolyzers (Maximize Market Research). Infrastructure also represents another challenge and logistics with the storage of green ammonia presents a significant financial hurdle. Lastly, uncertainty with government policies can create delays restricting the necessary investment for growth. (Topsoe, Proton Ventures)

Conclusion

This project examined the design of an integrated ammonia–urea production facility using the conventional Haber–Bosch process for ammonia synthesis and the Stamicarbon CO₂ stripping process for urea synthesis. The process begins with natural gas reforming to generate hydrogen, followed by ammonia synthesis from hydrogen and nitrogen, and finally urea synthesis by reacting

ammonia with recovered CO₂. The design highlights the importance of heat and steam integration, recycling unconverted reactants, and maximizing energy efficiency to make large-scale fertilizer production economical and sustainable.

In addition to the conventional design, emerging alternatives such as green hydrogen electrolysis and green ammonia were reviewed. These technologies offer a pathway to net-zero emissions by eliminating fossil fuel use but are currently limited by high energy demand and cost. As renewable electricity becomes cheaper and electrolyzer technology matures, these processes are likely to play a key role in decarbonizing the fertilizer industry.

Overall, this study demonstrates both the technical feasibility and global importance of ammonia and urea production, providing insight into how process selection, energy integration, and future innovation will shape the next generation of fertilizer plants.

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